From a Divergent Newton Series to a Convergent Interpolation of the Primes

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Abstract

We start from the Newton forward-difference series built on the leading entries of the difference table of the primes, $\sum_{n\geq 0}\Delta^n p_1\binom{x-1}{n}$, and show that it diverges at every non-integer x because $|\Delta^n p_1|$ grows like 2^n. We identify that growth rate as the signature of content at the Nyquist frequency, which places the prime sequence outside the convergence region of the Newton basis. We then track a sequence of repairs. Summation methods fail for a structural reason (a branch point on the Abel path). Recentring the window produces the first convergent form, with residual $O(h^{-1/2})$. Recasting the sliding windows in barycentric form leads to the Floater–Hormann family, and the blending degree $d = 1$ turns out to be exactly the endpoint correction that converts the divergent alternating series in the numerator into one whose partial sums settle, via the first step of Euler–Boole summation. Numerically the resulting object converges as the prime list is extended, and the approach to the limit is a single Nyquist mode $c_M \sin(\pi x)$; we show that this factorization is exact in a cardinal-series form of the interpolant. An exact Abel identity splits c_M into an analytic part, proved here to be $O(\log M/M)$ unconditionally, and an arithmetic part governed by the alternating sum of prime gaps $G(k) = \sum_{j\leq k}(-1)^j g_j$, bounded by $O(\Phi(M)/M)$ for any regular majorant Φ of $|G|$. The arithmetic part dominates numerically by two to three orders of magnitude, so $c_M = O(1/M)$ is false: it would force $G(k) = O(\log k)$, whereas $|G(2^{18})| = 1.23 \times 10^4$. This also explains quantitatively why Richardson extrapolation in $1/M$ gains about three digits and then stalls. We reduce the rate question to square-root cancellation in $\sum_{j\leq k}(-1)^j(g_j - \log p_j)$, equivalently in the alternating sum of the primes, and show that the Riemann Hypothesis controls the unsigned counterpart of this sum and nothing else. We prove that $G(k)$ is exactly the sum of $(-1)^{\pi(n)}$ over the integers, and that the existence of the limit forces $G(k) = o(k)$, so existence is itself an arithmetic statement. We measure G to be diffusive with no drift, with a long-run step variance about five percent above $\mathrm{Var}(g)$ because consecutive gaps are negatively correlated, and we find that the constant governing it matches the Hardy–Littlewood sieve models and not the Cramér model. Finally we prove two conditional results: under a diffusive hypothesis on G the limit exists, with root-mean-square rate $\log M/\sqrt{M}$; and under the Hardy–Littlewood conjecture, in Gallagher's uniform form, the parity correlations of π are Poisson on the scale $\log N$, so that the short-window variance constant tends to 1. All claims are accompanied by the numerical checks that support them, and Section 14 states plainly what is established and what is not.

1 The object

Let $p_1 = 2, p_2 = 3, p_3 = 5, \ldots$ and let Δ be the forward difference on the index, $\Delta p_k = p_{k+1} - p_k$. Writing $\Delta^n p_c$ for the c-th entry of the n-th row of the difference table, the Newton forward-difference series based at $c = 1$ is

$$N_h(x) = \sum_{n=0}^{h} \Delta^n p_1 \binom{x-1}{n}, \qquad \Delta^n p_1 = \sum_{j=0}^{n} (-1)^{n-j}\binom{n}{j} p_{j+1}. \tag{1}$$

At integers $x = 1, \ldots, h+1$ this reproduces p_x exactly, so it is a candidate interpolation of the primes to real argument. The first twenty leading differences are

$$2,\ 1,\ 1,\ -1,\ 3,\ -9,\ 23,\ -53,\ 115,\ -237,$$
$$457,\ -801,\ 1213,\ -1389,\ 445,\ 3667,\ -15081,\ 41335,\ -95059,\ 195769.$$

The question of this note is whether $h \to \infty$ in (1) produces anything, and if not, what does.

2 Why the series diverges

2.1 Growth of the leading differences

Computing the full difference table of the first 250 primes, the leading entries alternate in sign from $n = 3$ onward with only seven exceptional adjacent pairs, at $(14, 15)$, $(25, 26)$, $(38, 39)$, $(46, 47)$, $(97, 98)$, $(111, 112)$ and $(175, 176)$, and their magnitudes grow geometrically. The consecutive ratios over the last six rows are

$$2.0521, \quad 2.0483, \quad 2.0451, \quad 2.0426, \quad 2.0407, \quad 2.0393,$$

drifting slowly downward. The final entry $\Delta^{249} p_1$ has 75 digits. Removing the anomalous first gap by restarting the table at $p_2 = 3$ changes nothing: the corresponding ratios are $2.0447, \ldots, 2.0380$. So we take

$$|\Delta^n p_1| \ \sim\ C\rho^{\,n}, \qquad \rho \approx 2. \tag{2}$$

That $\rho \approx 2$ is not special to the primes. For a sequence with random-like content at the highest representable frequency, $\Delta^n f_1 = \sum_j (-1)^{n-j} \binom{n}{j} f_{j+1}$ has root-mean-square size

$$\Big(\textstyle\sum_j \binom{n}{j}^2 \Big)^{1/2} = \binom{2n}{n}^{1/2} \approx 2^n (\pi n)^{-1/4},$$

so 2^n is the generic ceiling. The primes sit essentially at it.

2.2 Asymptotics of the terms

Using the reflection identity

$$\binom{x-1}{n} = (-1)^n \binom{n-x}{n} = (-1)^n \frac{\Gamma(n-x+1)}{\Gamma(n+1)\,\Gamma(1-x)} \ \sim\ \frac{(-1)^n n^{-x}}{\Gamma(1-x)}, \tag{3}$$

and combining with (2), the n-th term of (1) behaves like

$$\Delta^n p_1 \binom{x-1}{n} \ \sim\ \frac{C\rho^n n^{-x}}{\Gamma(1-x)}. \tag{4}$$

The two sign alternations cancel. The terms are eventually of one sign and grow geometrically, so there is no cancellation available and the series diverges at every non-integer x.

This is confirmed directly. Using 250 leading differences, every term from $n = 200$ to $n = 205$ is positive at $x = 1.5$ and negative at $x = 2.5$ and $x = 10.5$, with consecutive ratios at $n = 249$ of 2.027, 2.019 and 1.953 respectively. Figure 1(a) shows the term magnitudes climbing on a straight line of slope $\log_{10} 2$.

Two structural facts fall out of (4). The factor $1/\Gamma(1-x)$ has poles at $x = 1, 2, 3, \ldots$, so all terms past $n = x - 1$ vanish identically at the integers, which is why the series is trivially exact there. And the sign of $1/\Gamma(1-x)$ flips on each successive unit interval, which is why the truncated sums blow up alternately upward and downward as x crosses each integer.

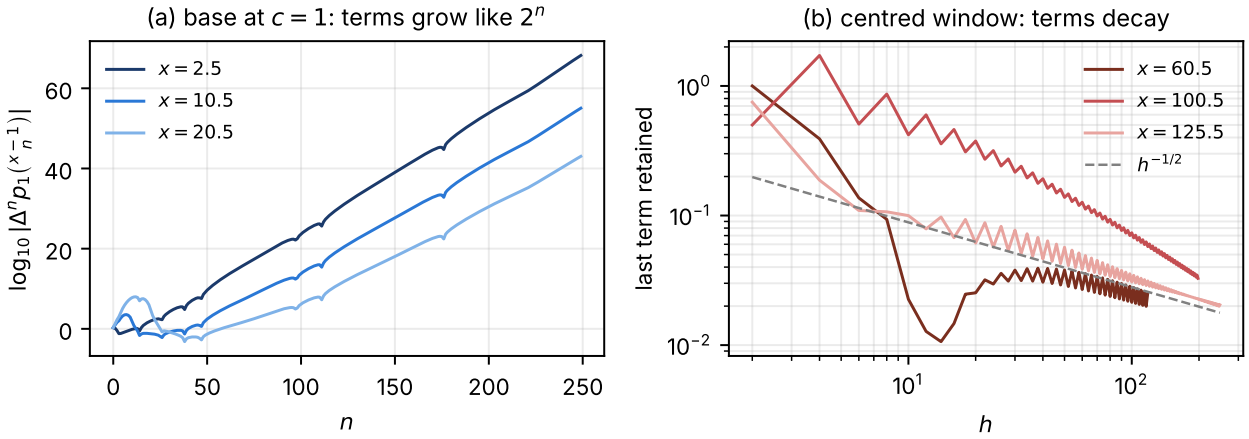


Figure 1: (a) Term magnitudes of (1) at three abscissas, with the base at $c = 1$: after an initial dip the terms grow geometrically at rate ≈ 2. (b) The last term retained by the centred scheme of Section 4, against window width h: the same data, reorganised, now decays like $h^{-1/2}$.

Remark 1. This is not Runge's phenomenon. Runge concerns a fixed analytic function sampled on equispaced nodes as the degree grows, and its standard cure is to cluster the nodes in a Chebyshev distribution. Here the nodes are the positive integers and cannot be moved, because they are where the data lives. The mechanism is the growth of the divided differences, not the geometry of a fixed interval.

2.3 The bandwidth reading

Growth rate ρ has a spectral meaning. Applying Δ^n to a mode $e^{i\omega k}$ gives $(e^{i\omega} - 1)^n$, of magnitude $|2\sin(\omega/2)|^n$. For a sequence band-limited to $|\omega| \le \tau$ the difference growth is therefore at most $2\sin(\tau/2)$, an increasing function of τ on $[0, \pi]$ with maximum value 2 attained exactly at the Nyquist frequency $\tau = \pi$. The measured $\rho \to 2$ says the prime sequence saturates the band.

The Newton basis, by contrast, converges only for data extending to a function of exponential type below $\ln 2 \approx 0.693$ (Nörlund). The gap is a factor of $\pi / \ln 2 \approx 4.5$. The divergence is therefore not an artefact of truncation order or of the base point; it is a mismatch between the bandwidth of the data and the bandwidth the basis can represent. Any repair must either change the basis or change what is meant by summation.

3 Two repairs that do not work

3.1 Summation methods

Since the terms are eventually geometric with ratio near 2, Shanks or Wynn acceleration ought to sum them the way it assigns -1 to $\sum 2^n$. It does not. On the clean model problem $f_k = (-1)^{k-1}$, whose leading differences are exactly $(-2)^n$ and whose natural continuation is $\cos(\pi(x-1))$, the ε-algorithm returns -5.6×10^{81} at $x = 1.5$ and -0.404 at $x = 2.25$ against the true -0.707.

The obstruction is structural rather than numerical. The generating function is

$$\sum_{n \ge 0} (-2)^n \binom{x-1}{n} z^n = (1 - 2z)^{x-1}, \tag{5}$$

which has a branch point at $z = \frac{1}{2}$. The Abel path from $z = 0$ to $z = 1$ runs into the cut. The two branches give $e^{\pm i\pi(x-1)}$, so no real Abel sum exists; the best available statement is that their average is $\cos(\pi(x-1))$. A real-valued summation method applied on the real axis is being asked for an object that is not there.

3.2 Optimal truncation

Treating (1) as an asymptotic series and stopping at the least term does work, in the sense of producing a bounded curve. The least term is 2.1×10^{-2} at $x = 4.5$ and 6.1×10^{-4} at $x = 19.5$, and the resulting values (9.815 at $x = 4.5$, 20.411 at $x = 8.5$, 39.596 at $x = 12.5$) sit sensibly between the bracketing primes. But the accuracy is capped by the least term and the curve jumps where the optimal order switches, by 1.62 at $x \approx 3.24$ and 0.70 at $x \approx 9.59$. It is a superasymptotic answer, not a convergent one.

4 Recentring: the first convergent form

Re-base the series at an arbitrary c,

$$N_{c,h}(x) = \sum_{n=0}^{h} \Delta^n p_c \binom{x-c}{n}, \tag{6}$$

which interpolates $p_c, \ldots, p_{c+h}$. A table built from N primes has $N - n$ entries in row n, so the c-th entry exists precisely when $n \le N - c$; the truncation bound $h \le N - c$ is forced by the data rather than chosen.

The crucial quantity is where x sits inside the window. Writing $x - c = h/2 + t$ with $t \in [0, 1)$,

$$\left|\binom{x-c}{h+1}\right| = \frac{1}{(h+1)!}\prod_{j=0}^{h}\left|\frac{h}{2} + t - j\right| \;\approx\; \frac{\left((h/2)!\right)^2}{(h+1)!} \;\sim\; \sqrt{\frac{\pi h}{2}}\,\frac{2^{-h}}{h}. \tag{7}$$

The 2^{-h} here cancels the ρ^h of (2) exactly in the limit $\rho \to 2$, leaving a residual of size $O(h^{-1/2})$. Centring converts divergence into convergence.

The cancellation is dramatic. At $x = 125.5$ with $c = 2$ and $h = 248$,

$$\Delta^{248} p_2 \cdot \binom{123.5}{248} = \left(1.828 \times 10^{74}\right) \times \left(1.119 \times 10^{-76}\right) = 0.020461,$$

about 150 orders of magnitude of cancellation landing on a small number. Figure 1(b) shows the last retained term following $h^{-1/2}$ once $h \gtrsim 30$. Converged values include 281.8389 at $x = 60.5$, 545.5067 at $x = 100.5$ and 695.8404 at $x = 125.5$.

Two defects remain. First, $O(h^{-1/2})$ is slow: with 250 primes one gets about two decimal places, and each further digit costs a hundredfold increase in h. Second, the construction is not canonical. If c is chosen per x, say $c = \mathrm{round}(x - h/2)$, the curve is discontinuous where c increments. That jump has a closed form: two adjacent windows agree at h common nodes, so their difference is $A\prod_{j=1}^{h}(x - c - j)$ with $A = \Delta^{h+1} p_c / h!$, giving at the switch point $x^* = c + \frac{h+1}{2}$

$$J = \frac{\Delta^{h+1} p_c}{h!}\prod_{j=1}^{h}\left(\frac{h+1}{2} - j\right). \tag{8}$$

Evaluated at $c = 60$ this reproduces the measured jumps exactly: 0.23724537 at $h = 20$, 0.15517069 at $h = 40$, -0.17773625 at $h = 80$, -0.061394627 at $h = 120$. Note that these are the same order as the truncation residual. The discontinuity is not an artefact of the switching rule; it is a faithful readout of how much the interpolant is undetermined.

5 From windows to weights

The natural way to remove the switching is to trust each window in inverse proportion to its own error factor, that is, to weight window i by the reciprocal of its node polynomial. Doing this for all sliding windows of a fixed length simultaneously is exactly the Floater–Hormann barycentric rational interpolant. With blending degree d and local polynomials P_i through nodes $x_i, \dots, x_{i+d}$,

$$r(x) = \frac{\sum_i \lambda_i(x) P_i(x)}{\sum_i \lambda_i(x)}, \qquad \lambda_i(x) = \frac{(-1)^i}{(x - x_i) \cdots (x - x_{i+d})}. \tag{9}$$

For equispaced nodes this collapses to the barycentric form

$$r_M(x) = \frac{\displaystyle\sum_{k=1}^{M} \frac{w_k}{x - k}\, p_k}{\displaystyle\sum_{k=1}^{M} \frac{w_k}{x - k}}, \qquad w_k = (-1)^k \sum_{j=\max(0,\, k+d-M)}^{\min(d,\, k-1)} \binom{d}{j}, \tag{10}$$

(the overall sign convention is immaterial, since it cancels in the quotient). This object is C^∞ with no poles on the real line, interpolates exactly at every node, and has a Lebesgue constant growing linearly rather than like 2^h, which removes the Runge behaviour outright.

For $d \le k \le M - d$ the inner sum is the full binomial total, so

$$w_k = (-1)^k 2^d \qquad \text{throughout the interior.} \tag{11}$$

Only the first d and last d weights depend on d at all. Numerically, $r(60.5)$ moves from 281.88161 at $d = 1$ to 281.89735 at $d = 12$, five significant figures of agreement, while the edge values move enormously: $r(2.5)$ runs from 4.238 at $d = 1$ to 25.43 at $d = 12$. The degree is an edge parameter.

Choosing $d = 1$ minimises overshoot past the bracketing primes (0.533, against 0.612 at $d = 0$, 17.50 at $d = 9$ and 145.03 at $d = 12$) and ties for lowest curvature. At $d = 1$ the weights are simply

$$w_k = (-1)^k \times \begin{cases} 1, & k = 1 \text{ or } k = M, \\ 2, & \text{otherwise.} \end{cases} \tag{12}$$

The endpoints are halved. Section 6 shows that this halving is not cosmetic.

6 Why $d = 1$ converts divergence into convergence

Dividing (12) through by 2, the numerator and denominator of (10) are

$$N_M(x) = \sum_{k=1}^{M} \frac{(-1)^k p_k}{x - k} - \tfrac{1}{2}\Big(\tfrac{-p_1}{x-1} + \tfrac{(-1)^M p_M}{x-M}\Big), \qquad D_M(x) = \sum_{k=1}^{M} \frac{(-1)^k}{x - k} - \tfrac{1}{2}\Big(\tfrac{-1}{x-1} + \tfrac{(-1)^M}{x-M}\Big). \tag{13}$$

6.1 The denominator converges, and in closed form

The terms of D_M decay like $1/k$ with alternating signs, so it converges. Splitting the Mittag-Leffler expansion $\pi/\sin(\pi x) = \sum_{k\in\mathbb{Z}}(-1)^k/(x-k)$ at $k=0$ gives

$$D_\infty(x) = \frac{\pi}{\sin \pi x} - \frac{1}{2}\Big[\psi\Big(\frac{x+1}{2}\Big) - \psi\Big(\frac{x}{2}\Big)\Big] + \frac{1}{2(x-1)}, \tag{14}$$

where the middle term is $\sum_{m\geq 0}(-1)^m/(x+m)$ and the last is the halved first weight. At $x = 60.5$ the three pieces are 3.14159265, -0.00833276 and $+0.00840336$, totalling 3.14166326, against the numerically summed 3.1416632601.

Expanding the last two pieces, $\frac{1}{2(x-1)} = \frac{1}{2x} + \frac{1}{2x^2} + O(x^{-3})$ and $\sum_{m\geq 0}\frac{(-1)^m}{x+m} = \frac{1}{2x} + \frac{1}{4x^2} + O(x^{-3})$, so they very nearly cancel and

$$D_\infty(x) = \frac{\pi}{\sin \pi x} + \frac{1}{4x^2} + O(x^{-3}). \tag{15}$$

At $x = 60.5$ the predicted correction is 6.8×10^{-5} and the observed one is 7.1×10^{-5}. Equation (15) is the key to Section 7.

6.2 The numerator diverges, and the halving repairs it

The numerator is the interesting one. Since $p_k \sim k \ln k$,

$$\frac{(-1)^k p_k}{x-k} \;\longrightarrow\; -(-1)^k \ln k, \tag{16}$$

so the terms do *not* tend to zero. As a series, $\sum_k (-1)^k p_k/(x-k)$ diverges outright. What rescues it is precisely the endpoint halving in (12).

For an alternating sum with slowly varying a_k, Euler–Boole summation gives

$$\sum_{k=1}^{M}(-1)^k a_k = C + \frac{(-1)^M a_M}{2} + \frac{(-1)^M a'_M}{4} + \cdots, \tag{17}$$

an asymptotic expansion whose leading oscillatory term is exactly half the last term. Subtracting that half term removes the leading oscillation and leaves a residual of size $O(a'_M)$. For a smooth sequence that would be $O(1/M)$; for the primes the first differences of a_k carry the gaps g_k, and Section 8 shows that this arithmetic contribution is the dominant one. Comparing (13) with (17), the $d = 1$ weight pattern is the first step of Euler–Boole summation, applied automatically by the interpolation scheme.

The numbers make this unambiguous. At $x = 60.5$, with $M = 10^3, 10^4, 10^5, 10^6$:

M	10^3	10^4	10^5	10^6
N_M with the halving	885.227	885.269	885.150	885.125
N_M without it	880.996	879.984	878.631	877.365

The first row settles; the second drifts without limit, losing about 1.3 per decade, which is the $\frac{1}{2}p_M/M \approx \frac{1}{2}\ln M$ growth predicted by (17). The gap between the rows at $M = 10^6$ is 7.7602, against the predicted half-term $\frac{1}{2}p_{10^6}/(10^6 - x) = \frac{1}{2}\cdot 15485863/999939.5 = 7.7434$, the small remainder being the halved first term $p_1/(2(x-1)) = 0.0168$.

This is the pivot of the whole construction. A divergent alternating series in the numerator is divided by a convergent one in the denominator, and a single endpoint correction, supplied for free by the choice $d = 1$, makes the partial sums of the numerator settle as well.

Remark 2. The choice $d = 0$, Berrut's original interpolant, does *not* halve the endpoints, so it does not carry the Euler–Boole correction. Correspondingly $r(60.5) = 281.620$ at $d = 0$ against 281.882 for every $d \geq 1$. Degrees $d \geq 2$ also halve the endpoints, in a more elaborate way, which is why they converge to the same interior limit. So $d = 1$ is the minimal choice that works, not the only one.

7 The residual is a single Nyquist mode

For $k \gg x$ one has $1/(x-k) = -1/k + O(x/k^2)$, so the tail contributions to N_M and D_M are independent of x to leading order. Writing $N_M = N_\infty + \delta_N$ and $D_M = D_\infty + \delta_D$ with δ_N, δ_D approximately x-free,

$$r_M(x) - r_\infty(x) \;\approx\; \frac{\delta_N - r_\infty \delta_D}{D_\infty(x)} \;\approx\; c_M \sin(\pi x), \tag{18}$$

the last step by (15). The prediction is sharp: the entire dependence on how many primes are used is one scalar times the Nyquist mode. Proposition 12 below shows that the factor $\sin(\pi x)$ is in fact exact, and only the remaining factor is approximately constant.

Solving (18) for c_M at eight abscissas confirms it. The reference r_∞ is the value computed from all 6.84×10^6 primes below 1.2×10^8:

x	60.15	60.25	60.5	60.75	60.85	61.5	124.3	125.5
c_M, $M = 10^4$	0.04644	0.04644	0.04644	0.04644	0.04644	0.04645	0.04660	0.04661
c_M, $M = 10^5$	0.00856	0.00856	0.00856	0.00856	0.00856	0.00856	0.00857	0.00857

The value is constant to five decimals across different integer cells, different fractional offsets, and both signs of $\sin(\pi x)$. The slight drift in the last two columns is the $1/(4x^2)$ term of (15), of the predicted size. The reference value is itself uncertain by about 1.1×10^{-3} (Section 13); because that error also has the Nyquist form, it shifts every entry of a row by the same amount, so the constancy across columns is unaffected while the absolute values carry that uncertainty.

Figure 2 shows $|c_M|$ decaying. The decay is irregular and no power law should be read off it by eye; Section 8 determines the rate properly. Whether the limit exists at all is an arithmetic question, not a numerical one: Proposition 21 shows that existence forces $G(k) = o(k)$, and Theorem 19 gives a sufficient condition.

Remark 3 (on a discarded estimate). An earlier version of these notes fitted $\log|c_M|$ against $\log M$ pointwise over $M \in [3 \times 10^3, 6 \times 10^5]$, obtained an exponent of -0.942, and read it as confirming $c_M = O(1/M)$. That estimator is not valid here. Section 8 shows that c_M contains a term behaving like a random walk, so individual values are not representative of their local scale and a least-squares line through them largely records where the walk happened to sit at the ends of the fitting window. Taking instead the root mean square of $|c_M|$ over dyadic blocks gives an exponent of -0.38, with block-to-block ratios ranging from 0.64 to 3.66. The honest reading of that is not "the exponent is -0.38" but "no exponent is determined by data over this range", which is what one should expect once the dominant term is arithmetic rather than analytic.

8 The rate: what can be proved

Throughout this section x is fixed and non-integral, $a_k = p_k/(x-k)$, $g_k = p_{k+1} - p_k$, and implied constants may depend on x.

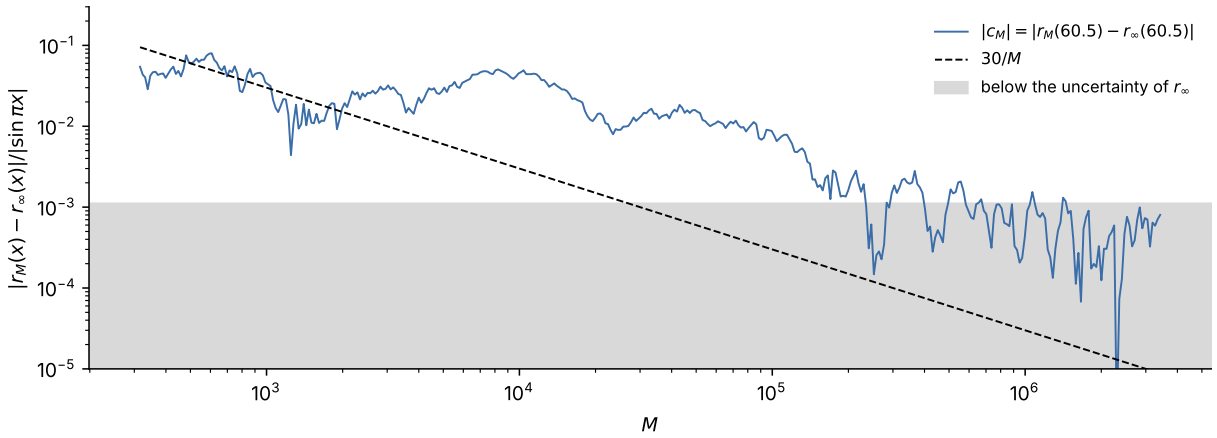


Figure 2: Amplitude of the residual Nyquist mode at $x = 60.5$ against the number of primes used, measured against r_∞ computed from 6.84×10^6 primes, with a $30/M$ reference. In the shaded region the reference value is not accurate enough to resolve c_M. The scatter is not measurement noise. Section 8 identifies it as an arithmetic term that dominates the smooth one.

8.1 An exact half-sum identity

Lemma 4. *For any sequence (a_k) and any $M \geq 2$,*

$$\sum_{k=1}^{M} (-1)^k a_k - \tfrac{1}{2}\Big((-1)a_1 + (-1)^M a_M\Big) \;=\; -\frac{1}{2}\sum_{k=1}^{M-1} (-1)^k \big(a_{k+1} - a_k\big). \tag{19}$$

Proof. Put $A_k = (-1)^k/2$, so that $A_k - A_{k-1} = (-1)^k$. Abel summation gives $\sum_{k=1}^{M}(A_k - A_{k-1})a_k = A_M a_M - A_0 a_1 - \sum_{k=1}^{M-1} A_k(a_{k+1} - a_k)$. Substituting $A_M = (-1)^M/2$ and $A_0 = \frac{1}{2}$ and moving the two boundary terms to the left gives (19). □

The left side of (19) is exactly the numerator $N_M(x)$ of (13). So the $d = 1$ scheme does not merely approximate an Euler–Boole correction; it is identically equal to a half-sum of first differences. Checked against direct evaluation at $x = 60.5$, the two sides agree to ten decimals at $M = 10, 10^2, \ldots, 10^6$, for instance 885.1252016789 against 885.1252016788 at $M = 10^6$. Consequently, whenever the series converges,

$$N_\infty(x) - N_M(x) = -\frac{1}{2}\sum_{k \geq M} (-1)^k \Delta a_k, \tag{20}$$

and the whole question of the rate, and of existence, is the size of that tail.

8.2 Splitting the tail

A direct computation gives the exact decomposition

$$\Delta a_k = \underbrace{\frac{g_k}{x - k - 1}}_{\text{arithmetic}} + \underbrace{\frac{p_k}{(x - k - 1)(x - k)}}_{\text{analytic}}, \tag{21}$$

verified numerically to 5.7×10^{-14} over $k \le 10^6$. The names are chosen because the first term carries the irregularity of the prime gaps while the second depends on p_k only through its smooth size.

Proposition 5 (analytic part)**.** *Let* $\phi_k = p_k/\big((k+1-x)(k-x)\big)$*. Then* $\sum_k (-1)^k \phi_k$ *converges and* $\big|\sum_{k\ge M}(-1)^k\phi_k\big| \ll_x \log M/M$*.*

Proof. Since the partial sums of $(-1)^k$ are bounded by 1 and $\phi_k \to 0$, Abel summation bounds the tail by the total variation $\sum_{k\ge M}|\phi_{k+1}-\phi_k|$, and convergence follows once that is finite. Clearing denominators,

$$\phi_{k+1}-\phi_k = \frac{g_k(k-x)-2p_k}{(k+2-x)(k+1-x)(k-x)} \ll \frac{g_k}{k^2}+\frac{p_k}{k^3}.$$

For the second term, $p_k \ll k\log k$ gives $\sum_{k\ge M} p_k/k^3 \ll \sum_{k\ge M}\log k/k^2 \ll \log M/M$. For the first, partial summation with $\sum_{k\le K} g_k = p_{K+1}-p_1 \ll K\log K$ gives

$$\sum_{k\ge M}\frac{g_k}{k^2} \ll \frac{M\log M}{M^2}+\sum_{k\ge M}\frac{k\log k}{k^3} \ll \frac{\log M}{M}. \qquad \square$$

Only Chebyshev-type bounds are used, so Proposition 5 is unconditional.

Proposition 6 (arithmetic part)**.** *Let* $G(k)=\sum_{j\le k}(-1)^j g_j$ *and suppose* Φ *is nondecreasing with* $|G(k)| \le \Phi(k)$ *and* $\Phi(k) = k^\theta L(k)$ *for some* $\theta < 1$ *and slowly varying* L*. Then the series below converges and*

$$\Big|\sum_{k\ge M}\frac{(-1)^k g_k}{k+1-x}\Big| \ll_x \frac{\Phi(M)}{M}.$$

Proof. Write $\psi_k = (k+1-x)^{-1}$, which decreases to 0 with $\psi_k - \psi_{k+1} \ll k^{-2}$. Abel summation against the partial sums $G(k)-G(M-1)$ gives

$$\Big|\sum_{k\ge M}(-1)^k g_k\psi_k\Big| \le 2\Phi(M)\psi_M + 2\sum_{k\ge M}\Phi(k)(\psi_k-\psi_{k+1}) \ll \frac{\Phi(M)}{M}+\sum_{k\ge M}\frac{\Phi(k)}{k^2},$$

and $\theta<1$ makes the last sum $\ll \Phi(M)/M$. $\square$

Proposition 7. *With* Φ *as above,* $|c_M| \ll_x \dfrac{\log M}{M}+\dfrac{\Phi(M)}{M}$*.*

Proof. Combine (20), (21), Propositions 5 and 6, and divide by $D_\infty(x)$, which is bounded away from 0 for fixed non-integral x by (15). The corresponding tail for D_M is the special case $p_k \equiv 1$ of (21), for which the arithmetic term is absent and Proposition 5 applies directly. $\square$

8.3 Why $c_M = O(1/M)$ is false

Proposition 7 reduces the rate to the growth of the alternating gap sum G. Reading it backwards: $c_M = O(1/M)$ would require $\Phi(M) = O(1)$, or with the $\log M/M$ term absorbed, $G(k) = O(\log k)$. That is essentially total cancellation in $\sum_{j\le k}(-1)^j g_j$, and it is contradicted by the data. Figure 3(b) plots $|G(k)|$: it reaches 1.23×10^4 at $k = 2^{18}$, against $\log k \approx 12.5$. The root mean square of $|G|$ over dyadic blocks tracks $\sqrt{k}\log k$ within a factor of two, drifting from 0.38 times that reference at $k \approx 2^8$ to 1.56 at $k \approx 2^{18}$, which is the behaviour of a quantity with square-root cancellation sitting in a long one-signed excursion, not of a bounded one.

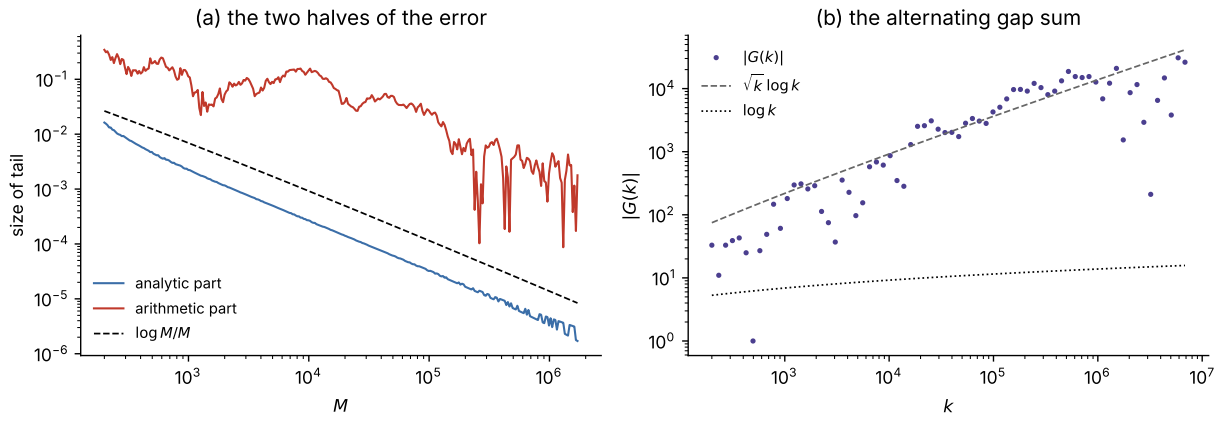


Figure 3: (a) The two tails of (20) at $x = 60.5$, using all primes below 1.2×10^8. The analytic part follows $\log M/M$; the arithmetic part is far larger and irregular. (b) The alternating gap sum $|G(k)|$, against $\sqrt{k}\log k$ and against $\log k$. Bounding $|G|$ by $\log k$ is what $c_M = O(1/M)$ would require.

Corollary 8. *If the alternating gap sum exhibits square-root cancellation, $G(k) \ll k^{1/2+\varepsilon}$, then $c_M \ll_{x,\varepsilon} M^{-1/2+\varepsilon}$.*

That is the strongest clean statement available, and it is conditional. Unconditionally, the trivial bound $|G(k)| \le p_{k+1}$ gives only $c_M \ll \log M$, which says nothing; in particular it does not show that $c_M \to 0$.

The split is also directly visible. Evaluating the two tails of (20) with (21) separately at $x = 60.5$, using the 1 031 130 primes below 1.6×10^7:

M	analytic tail	$\log M/M$	arithmetic tail
10^3	-0.002258	0.006908	-0.098696
10^4	-0.000265	0.000921	-0.142631
10^5	-0.000029	0.000115	-0.023858
3×10^5	-0.000008	0.000042	-0.001165

The analytic tail sits between a third and a fifth of $\log M/M$ and decays cleanly, as Proposition 5 predicts. The arithmetic tail is two to three orders of magnitude larger and does not decay monotonically at all. Whatever c_M does over this range, it is doing it for arithmetic reasons.

9 Targeting a bound on G

Proposition 7 makes the rate of c_M a question about $G(k) = \sum_{j\le k}(-1)^j g_j$ alone. This section reduces that question to a sharper one, records what the Riemann Hypothesis does and does not contribute, gives the measured behaviour, and compares it with the random models behind the Hardy–Littlewood conjecture.

9.1 Reduction to gap fluctuations

Proposition 9. $\left|\sum_{j\le k}(-1)^j \log p_j\right| \le \log p_k$ *for all* $k \ge 1$.

Proof. Apply Lemma 4 with $a_j = \log p_j$. The right side of (19) is bounded by $\frac{1}{2}\sum_{j<k} |\log p_{j+1} - \log p_j|$, which telescopes to $\frac{1}{2}(\log p_k - \log p_1)$ because $\log p_j$ increases. Restoring the two boundary terms, each of modulus at most $\frac{1}{2}\log p_k$, gives the claim. □

Writing $\varepsilon_j = g_j - \log p_j$ for the fluctuation of the gap about its mean size, this gives the unconditional reduction

$$G(k) = \sum_{j\le k} (-1)^j \varepsilon_j + O(\log k). \tag{22}$$

Numerically the error term is not merely bounded by $\log p_k$ but sits at about half of it: at $k = 10^2, 10^3, \dots, 10^6$ it takes the values $2.60, 3.97, 5.26, 6.53, 7.77$ against $\log p_k = 6.2, 8.9, 11.5, 14.1, 16.6$. So the smooth part of the gap contributes nothing, and the entire difficulty is the signed sum of the fluctuations.

9.2 Equivalence with the alternating sum of primes

Writing $\Sigma(k) = \sum_{i\le k} (-1)^i p_i$ and reindexing,

$$G(k) = -2\Sigma(k) + (-1)^k p_{k+1} - 2, \tag{23}$$

so a bound on G is equivalent to one on the alternating sum of the primes themselves, and Lemma 4 identifies $G(k-1)$ with -2 times the half-endpoint-corrected $\Sigma(k)$. The same phenomenon therefore appears twice in these notes: the interpolation error is a half-sum of gaps, and the gap sum is a half-sum of primes.

9.3 What the Riemann Hypothesis gives, and what it does not

The unsigned sum of fluctuations is completely standard. The gaps telescope, $\sum_{j\le k} g_j = p_{k+1} - p_1$, and $\sum_{j\le k} \log p_j = \vartheta(p_k)$, so

$$\sum_{j\le k} \varepsilon_j = p_{k+1} - \vartheta(p_k) - 2,$$

and on RH, using $\vartheta(x) = x + O(\sqrt{x}\log^2 x)$,

$$\sum_{j\le k} \varepsilon_j \ll \sqrt{p_k}\log^2 p_k \ll \sqrt{k\log k}\,\log^2 k. \tag{24}$$

At $k = 10^6$ the left side is 6.03×10^3 against a bound of 1.06×10^6, so this is far from tight, but it is unconditional in form and cheap.

It is also useless for our purpose. Splitting $A = \sum_{j\le k,\ j \text{ even}} \varepsilon_j$ and $B = \sum_{j\le k,\ j \text{ odd}} \varepsilon_j$, equation (24) controls $A + B$ while (22) asks for $A - B$. These are independent pieces of information, and no bound on the one constrains the other.

The obstruction can be stated more sharply. Since $(-1)^j = (-1)^{\pi(p_j)}$, the sum in (22) is a sum over primes weighted by the parity of the prime-counting function. Analytic methods reach π through ζ, which determines its size but carries no information about its parity, and there is no Dirichlet series with $(-1)^{\pi(p)}$ for coefficients that is amenable to the usual machinery. That is why the trivial bound $|G(k)| \le p_{k+1} - 2 \ll k\log k$ has not been improved here. Corollary 15 below puts this in its sharpest form.

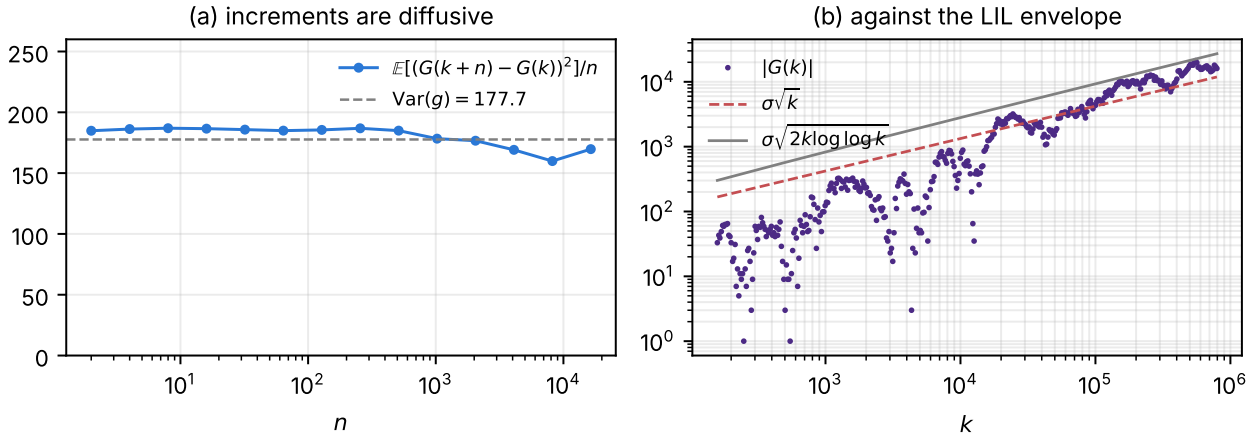


Figure 4: (a) Mean square increment of G per step, against step size, over $k \in [2 \times 10^5, 10^6]$. Flat means diffusive. The dashed line is the measured gap variance. (b) $|G(k)|$ against $\sigma\sqrt{k}$ and against the LIL envelope $\sigma\sqrt{2k \log\log k}$.

9.4 The diffusive model, measured

If consecutive gap fluctuations are uncorrelated, G is a mean-zero random walk with step variance $\mathrm{Var}(g)$, so $\mathrm{Var}(G(k)) \sim k\,\mathrm{Var}(g)$. This is testable without any modelling assumption, by measuring increments.

Over $k \in [2 \times 10^5, 10^6]$, where $\mathrm{Var}(g) = 177.66$:

n	1	16	64	256	1024	16384
$\mathbb{E}\big[(G(k+n)-G(k))^2\big]/n$	431.1	186.5	184.9	186.8	178.3	169.7

The ratio is flat across four orders of magnitude in n, which is diffusive behaviour with no drift term; a linear drift would make the ratio grow like n. The $n = 1$ entry is not an exception: a single increment is $\pm g_{k+1}$, so its mean square is $\mathbb{E}[g^2] = \mathrm{Var}(g) + \mathbb{E}[g]^2 = 177.66 + 253.44 = 431.10$, which is the tabulated value to five figures. For even n the mean cancels and the ratio drops to the long-run step variance of the walk, which Section 9.6 shows is about five percent above $\mathrm{Var}(g)$ because consecutive gaps are negatively correlated.

Two further checks. The mean increment over that range is 0.00362 ± 0.02321, consistent with zero, and the mean gap at even and odd index agrees to four decimals (15.9233 against 15.9160), so there is no parity bias at the level the data can resolve. And Figure 4(b) shows $|G(k)|$ staying inside the law-of-the-iterated-logarithm envelope $\sigma\sqrt{2k \log\log k}$, at a ratio rising from 0.13 at $k = 2^{12}$ to 0.83 at $k = 2^{19}$, which is a walk in a long one-signed excursion rather than a quantity with a trend.

One incidental observation: $\mathrm{Var}(g)/\log^2 p_k = 177.66/253.44 = 0.70$ on this range, so the gaps are underdispersed relative to the exponential distribution that Cramér's model predicts. That is consistent with the known failure of the naive Cramér model, and it makes the walk slightly slower than a first guess would suggest, but it does not change the exponent.

9.5 Conditional bound

Corollary 10. *If G obeys a law of the iterated logarithm with step variance $\sigma^2 = S\log^2 p_k \asymp \log^2 k$, where S is given by (25), then $G(k) \ll \sqrt{k}\,\log k\,\sqrt{\log\log k}$, and hence by Proposition 7,*

$$c_M \;\ll_x\; \frac{\log M\sqrt{\log\log M}}{\sqrt{M}}.$$

This is the target. It is consistent with every measurement above and it is what one should try to prove. What would suffice is square-root cancellation in $\sum_{j\le k}(-1)^j\varepsilon_j$, which by (22) is the whole problem, and which by (23) is equivalent to square-root cancellation in the alternating sum of the primes. As far as these notes are aware that is open, and the parity-of-π obstruction above suggests it is not a small gap.

9.6 Gap correlations and Hardy–Littlewood models

The diffusive picture above treats consecutive gap fluctuations as uncorrelated. They are not, and the correction is small but measurable. This subsection fixes the step variance of the walk, compares it with the random models behind the Hardy–Littlewood prime k-tuple conjecture, and uses the result to predict the size of the arithmetic tail directly. Everything here is heuristic or numerical; Section 11 proves what can be proved.

The step variance. Write $\hat g_k = g_k/\log p_k$ for the normalized gaps. Over $k \in [2\times 10^5, 10^6]$ their autocovariances at lags $1,\dots,8$ are

$$-0.0292,\ -0.0145,\ -0.0109,\ -0.0076,\ -0.0057,\ -0.0059,\ -0.0037,\ -0.0046,$$

against a variance of 0.6996. All are negative. For an alternating sum, odd lags then add to the variance and even lags subtract from it, and the odd lags win. The quantity that governs G is therefore not $\mathrm{Var}(\hat g)$ but the spectral density of $\hat g$ at frequency π,

$$S = \sum_{\ell\in\mathbb{Z}} (-1)^\ell \gamma(\ell), \tag{25}$$

where γ is the autocovariance of $\hat g$. The long-run increment ratio of the table above should be $S\log^2 p_k$, not $\mathrm{Var}(g)$. Repeating that measurement with non-overlapping increments, so that standard errors can be attached, gives

n	16	64	256	1024	4096	16384
$\mathbb{E}\big[(G(k+n)-G(k))^2\big]/n$	186.4	185.3	187.6	185.5	152.8	177.4
standard error	1.2	2.3	4.6	9.3	17.6	43.3

The plateau is about 186, which is $\mathrm{Var}(g) = 177.66$ plus five percent, and at $n = 16$ the difference is about seven standard errors. Summing (25) over lags up to 8 gives $S \approx 0.734$ on this range, so $S\log^2 p_k \approx 0.734 \times 253.5 \approx 186.0$, in agreement. The walk is still diffusive and the exponent in Corollary 10 is unchanged; only its step variance is $S\log^2 p_k$ rather than $\mathrm{Var}(g)$.

Comparison with Hardy–Littlewood models. Besides the Cramér model, we use the refined models that first remove the multiples of the primes up to some bound and then keep each remaining integer m independently with probability $\frac{z}{\varphi(z)} \cdot \frac{1}{\log m}$, where z is the product of the removed primes. As the bound grows these models reproduce the singular series of Hardy and Littlewood [4], and they are the standard heuristic for local prime statistics, as in Gallagher's derivation of Poisson behaviour in short intervals [3] and the consecutive-prime biases of Lemke Oliver and Soundararajan [5]. All of them satisfy the stationarity behind (25), so each has a well-defined S.

We estimated S by splitting the gap sequence into blocks of 1000 consecutive gaps, forming the alternating sum over each block, and dividing its mean square by 1000 times the block average of $\log^2 p_j$. For the primes up to 6×10^7 (3.56×10^6 primes) this gives $S = 0.763 \pm 0.018$. The two halves of the range give 0.728 and 0.798, which agree at that error level. Two independent runs of each model over the same range give:

model	S	$\operatorname{Var} \hat{g}$	lag-one correlation of $\hat{g}$
Cramér (no sieve)	0.937	0.94	$+0.000$
sieve by 2	0.881	0.88	-0.000
sieve by $2, 3$	0.793	0.82	-0.002
sieve by $2, 3, 5$	0.786	0.79	-0.008
sieve by $2, \ldots, 7$	0.792	0.76	-0.013
sieve by $2, \ldots, 11$	0.793	0.75	-0.018
sieve by $2, \ldots, 13$	0.778	0.74	-0.021
primes	0.763	0.71	-0.039

Each model value has a standard error of about 0.014. The Cramér value lies about seven standard errors above the primes; this is the same underdispersion noted above, now measured at the frequency that matters for G. Once the multiples of 2 and 3 are removed, every model agrees with the primes to within about one to two standard errors. The variance and the lag-one correlation keep moving toward the prime values as the sieve grows, reflecting the slow convergence of the singular series, but their effects on S offset, so at this height S settles early in the sieve. Figure 5(b) shows the comparison.

S is not a fixed constant of the primes. Corollary 18 shows that, under the Hardy–Littlewood conjecture, the corresponding short-window constant tends to 1 as the height grows. The values above are finite-height values, and across the heights tested they rise slowly, with fluctuations, from 0.708 for $n \in [10^5, 2 \times 10^5]$ to 0.762 for $n \in [3 \times 10^7, 6 \times 10^7]$ in the parity form of Section 10.

What this predicts for the tail. Write

$$T_M(x) = -\frac{1}{2} \sum_{k \geq M} (-1)^k \frac{g_k}{x - k - 1}$$

for the arithmetic part of (20). The weights $(k + 1 - x)^{-1}$ and $\log p_k$ vary slowly in k, so under the stationarity behind (25) only the spectral density at π contributes to its variance:

$$\sigma_M^2 := \operatorname{Var} T_M(x) \;\approx\; \frac{S}{4} \sum_{k \geq M} \frac{\log^2 p_k}{(k + 1 - x)^2} \;\sim\; \frac{S}{4} \frac{\log^2 M}{M}. \tag{26}$$

The mean of T_M comes from the smooth part $\log p_k$ of the gaps, is $O(\log M / M)$, and is smaller than σ_M. So the arithmetic tail is typically of size $\frac{1}{2}\sqrt{S} \log M / \sqrt{M}$, and after dividing by $D_\infty(x)$

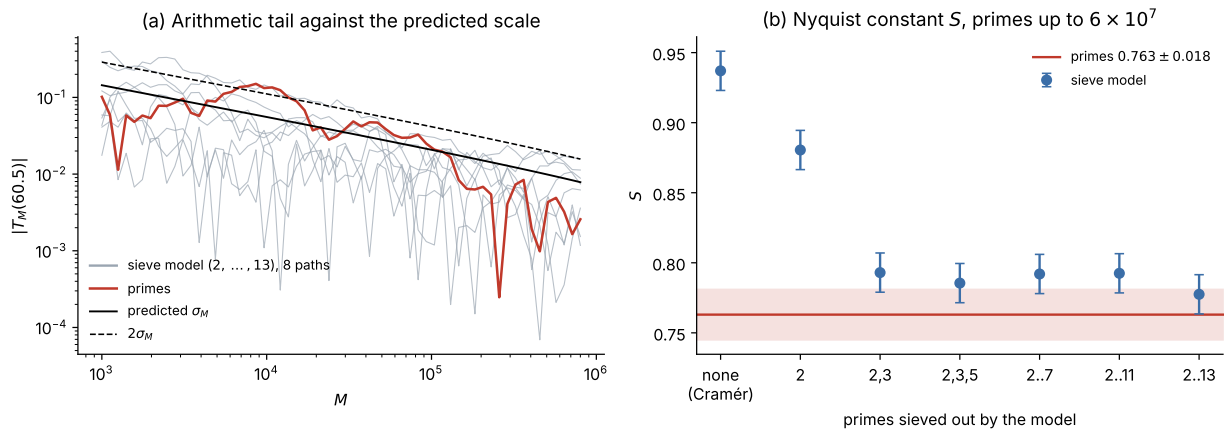


Figure 5: (a) The arithmetic tail $|T_M(60.5)|$ for the primes, with eight paths of the model sieved by $2, \ldots, 13$ and the predicted scale σ_M of (26). (b) The constant S of (25) for the primes up to 6×10^7, with its standard error band, against the sieve models.

as in Proposition 7, c_M is typically of order $M^{-1/2} \log M$, with the extra $\sqrt{\log \log M}$ of Corollary 10 appearing only in the worst case. Theorem 19 proves the mean-square form of this under a stated hypothesis.

We tested this at $x = 60.5$ using all primes up to 1.2×10^8 (6.84×10^6 primes), truncating the sum at the last available gap and computing σ_M from (26) with $S = 0.763$ and the same truncation:

M	$T_M(60.5)$	σ_M	T_M/σ_M
10^3	-0.10170	0.14422	-0.71
10^4	-0.14564	0.05553	-2.62
10^5	-0.02686	0.02073	-1.30
3×10^5	-0.00417	0.01263	-0.33

These differ slightly from the table in Section 8 because more primes are used in the tail. Over 60 values of M between 10^3 and 8×10^5, the root mean square of T_M/σ_M is 1.20. For comparison, 16 independent paths of the Cramér model and 16 of the model sieved by $2, \ldots, 13$ give root-mean-square ratios in $[0.60, 1.35]$ and $[0.53, 1.71]$, with medians 0.91 and 0.87, so the primes are inside both ranges. The prime tail exceeds $2\sigma_M$ only for $6\,800 \lesssim M \lesssim 13\,500$, reaching about $2.6\sigma_M$ near $M = 10^4$. Values of M this close together are strongly correlated, so this is one excursion; six of the sixteen sieved paths also exceed $2\sigma_M$ somewhere in the range. A power-law fit to the root mean square of T_M over dyadic blocks gives an exponent of -0.55 for the primes, against -0.43 for σ_M itself, which differs from $-1/2$ because of the factor $\log M$ over this short range. Single sieved paths give exponents from -0.73 to -0.22, with median -0.40, and the primes fall at the low end, one of the sixteen paths being lower. Figure 5(a) shows the prime tail among the model paths.

Scope. Shown numerically: the long-run step variance of G exceeds $\mathrm{Var}(g)$ by about five percent because of negative gap correlations; the constant S that measures this agrees with the Hardy–Littlewood sieve models once multiples of 2 and 3 are removed, and disagrees with the plain Cramér model by about seven standard errors; and the arithmetic tail has the size (26) predicts, with fluctuations typical of single model paths. Not shown: that the primes satisfy the stationarity behind (25), which is the same square-root cancellation that Corollary 10 assumes and is open;

an exponent for c_M sharper than about ± 0.25, which a single path over less than three decades of M cannot give; and a closed form for S in terms of the singular series, which would need the correlations in prime index rather than in position. By itself the tail does not separate the Cramér model from the sieved ones; that separation comes from S.

10 Three exact identities

This section records three identities. Each is elementary, and each was checked numerically against direct evaluation. Together they say what r_∞ is as an interpolant, why the approach to the limit is a single Nyquist mode, and how the alternating gap sum G is a sum over integers rather than over prime indices. Throughout, x is non-integral, $\operatorname{sinc} t = \sin(\pi t)/(\pi t)$, and

$$r_M(x) = \frac{\sum_{k=1}^{M} w_k^{(M)} (-1)^k \, p_k/(x-k)}{\sum_{k=1}^{M} w_k^{(M)} (-1)^k/(x-k)}, \qquad w_k^{(M)} = \begin{cases} \tfrac{1}{2}, & k = 1 \text{ or } k = M, \\ 1, & \text{otherwise,} \end{cases}$$

which is (13).

10.1 The cardinal-series form

Proposition 11. *For every $M \geq 2$ and non-integral x,*

$$r_M(x) = \frac{C_M(x)}{D_M^{\mathrm{c}}(x)}, \qquad C_M(x) = \sum_{k=1}^{M} w_k^{(M)} p_k \operatorname{sinc}(x-k), \qquad D_M^{\mathrm{c}}(x) = \sum_{k=1}^{M} w_k^{(M)} \operatorname{sinc}(x-k).$$

Proof. Since $\sin(\pi(x-k)) = (-1)^k \sin(\pi x)$, we have $\operatorname{sinc}(x-k) = \frac{\sin \pi x}{\pi} \cdot \frac{(-1)^k}{x-k}$. Multiplying the numerator and the denominator of r_M by the same non-zero factor $\sin(\pi x)/\pi$ gives the claim. □

So r_M is the Whittaker cardinal series of the primes, with the two end terms halved, divided by the cardinal series of the constant 1 with the same weights. Because $\sum_{k \in \mathbb{Z}} \operatorname{sinc}(x-k) = 1$, the limiting denominator is

$$D_\infty^{\mathrm{c}}(x) = 1 - \sum_{k \leq 0} \operatorname{sinc}(x-k) - \tfrac{1}{2} \operatorname{sinc}(x-1) = 1 + O(1/x),$$

so for large x the limit r_∞ is, up to a normalization tending to 1, the cardinal series of the primes in the index variable. This places the construction next to band-limited interpolation. The cardinal series of a sequence growing like $k \log k$ does not converge in the ordinary sense; halving the last term of each partial sum is what makes the partial sums settle, which is the content of Section 6 in this language.

10.2 The Nyquist factor is exact

Proposition 12. *Suppose $C_M(x)$ converges as $M \to \infty$. Define*

$$\alpha_M(x) = \sum_{k>M} \frac{(-1)^k p_k}{x-k} + \frac{(-1)^M p_M}{2(x-M)}, \qquad \beta_M(x) = \sum_{k>M} \frac{(-1)^k}{x-k} + \frac{(-1)^M}{2(x-M)}.$$

Then, exactly,

$$r_M(x) - r_\infty(x) = \frac{\sin \pi x}{\pi} \cdot \frac{r_\infty(x)\, \beta_M(x) - \alpha_M(x)}{D_M^{\mathrm{c}}(x)}.$$

Proof. Passing from M to ∞ changes the weight of $k = M$ from $\frac{1}{2}$ to 1 and adds the terms $k > M$, so $C_\infty - C_M = \sum_{k>M} p_k \operatorname{sinc}(x-k) + \frac{1}{2} p_M \operatorname{sinc}(x-M) = \frac{\sin \pi x}{\pi} \alpha_M$, and likewise $D^{\mathrm{c}}_\infty - D^{\mathrm{c}}_M = \frac{\sin \pi x}{\pi} \beta_M$. Writing $s = \sin(\pi x)/\pi$,

$$\frac{C_M}{D^{\mathrm{c}}_M} - \frac{C_\infty}{D^{\mathrm{c}}_\infty} = \frac{(C_\infty - s\alpha_M) D^{\mathrm{c}}_\infty - C_\infty (D^{\mathrm{c}}_\infty - s\beta_M)}{D^{\mathrm{c}}_M D^{\mathrm{c}}_\infty} = s \, \frac{r_\infty \beta_M - \alpha_M}{D^{\mathrm{c}}_M}. \qquad \square$$

The functions α_M and β_M depend on x only through $1/(x-k)$ with $k \geq M$, so for x much smaller than M they vary slowly in x. All the fast oscillation of the error is in the factor $\sin \pi x$. The description (18) is the approximation obtained by freezing the slowly varying factor. At $x = 60.37$, with 1.5×10^5 primes standing in for the limit, the two sides of Proposition 12 agree to about 10^{-9} relative for $M = 10^3$ and $M = 10^4$.

Remark 13. Solving Proposition 12 for r_∞ gives $r_\infty = (C_M + s\alpha_M)/(D^{\mathrm{c}}_M + s\beta_M) = C_\infty / D^{\mathrm{c}}_\infty$, so it does not yield a closed form: all the information about primes beyond M sits in α_M. The denominator correction β_M is explicit, and replacing p_k for $k > M$ by a smooth approximation (Cipolla's expansion, shifted to match the last known primes) completes the analytic part of α_M. Tested at $x = 60.5$ against the value from 6.84×10^6 primes, this tail completion changes the error of r_M only from 0.0331 to 0.0301 at $M = 10^3$ and from 0.0086 to 0.0084 at $M = 10^5$, because the error is dominated by the arithmetic tail, which no smooth model contains.

10.3 Gap sums are exponential sums of π

Proposition 14. *For every real θ and every $k \geq 1$,*

$$\sum_{j=1}^{k} g_j \, e^{ij\theta} = \sum_{2 \leq n < p_{k+1}} e^{i\theta\pi(n)}.$$

Proof. The integers $2 \leq n < p_{k+1}$ split into the blocks $p_j \leq n < p_{j+1}$, $j = 1, \ldots, k$. Block j contains g_j integers, and $\pi(n) = j$ on all of them. $\square$

The left side is the discrete Fourier transform of the gap sequence in the index variable, so the whole index spectrum of the gaps is a family of exponential sums of $\pi(n)$ over ordinary integers. Taking $\theta = \pi$ gives the case used throughout this paper.

Corollary 15. *$G(k) = \sum_{2 \leq n < p_{k+1}} (-1)^{\pi(n)}$. In particular, with $\psi_k = (k+1-x)^{-1}$, the arithmetic tail $T_M(x)$ of Section 9.6 is*

$$T_M(x) = \frac{1}{2} \sum_{k \geq M} (-1)^k g_k \, \psi_k = \frac{1}{2} \sum_{n \geq p_M} \frac{(-1)^{\pi(n)}}{\pi(n) + 1 - x}, \tag{27}$$

whenever either side converges.

Proof. The first statement is Proposition 14 with $\theta = \pi$. For the second, the weight $1/(\pi(n)+1-x)$ is constant on each block, equal to ψ_j, so the same block decomposition applies term by term to every partial sum ending at a prime. $\square$

Corollary 15 restates the obstruction of Section 9 in its sharpest form: everything about the rate of convergence, and as shown below about the existence of r_∞, is a statement about the parity of $\pi(n)$ along the integers. At $x = 60.5$ and $M = 5000$, the two sides of (27), truncated at the same prime, agree to all sixteen printed digits.

11 Conditional consequences of the Hardy–Littlewood conjecture

This section proves three things. First, the Hardy–Littlewood conjecture, in the uniform form used by Gallagher [3], implies that the parity correlations of π are asymptotically Poisson on the scale $\log N$; as a consequence the short-window variance of $\sum(-1)^{\pi(n)}$ has an explicit limit, which tends to 1 as the window grows, in contrast to the finite-height values of S in Section 9.6. Second, a diffusive hypothesis on G, of the kind measured in Section 9, implies that r_∞ exists, with a mean-square rate matching the heuristic and an explicit pointwise rate. Third, conversely, the existence of r_∞ at a single point already forces $G(k) = o(k)$. The link between the first and second results is not closed here, and the end of the section says exactly what is missing.

11.1 Hardy–Littlewood implies Poisson parity correlations

For a finite set D of integers, let $\mathfrak{S}(D) = \prod_p \big(1-\nu_D(p)/p\big)(1-1/p)^{-|D|}$, where $\nu_D(p)$ is the number of residues modulo p occupied by D. Write $L = \log N$, $\mathbf{1}_{\mathcal{P}}$ for the indicator of the primes, and, for $N < n \le 2N$,

$$\mathcal{N}(n,h) = \pi(n+h) - \pi(n), \qquad C_N(h) = \frac{1}{N}\sum_{N<n\le 2N} (-1)^{\mathcal{N}(n,h)} = \frac{1}{N}\sum_{N<n\le 2N} (-1)^{\pi(n)}(-1)^{\pi(n+h)}.$$

Hypothesis HL**.** For every integer $k \ge 1$ and every $\mu > 0$,

$$\sup_{\substack{D\subset[1,\mu L]\\ |D|=k}} \Big| L^k \cdot \frac{1}{N}\sum_{N<n\le 2N}\prod_{d\in D}\mathbf{1}_{\mathcal{P}}(n+d) - \mathfrak{S}(D)\Big| \longrightarrow 0 \qquad (N\to\infty).$$

This is the prime k-tuple conjecture of Hardy and Littlewood [4] for each fixed k, with the error uniform over tuples of diameter $O(\log N)$, which is the form Gallagher used. The factor L^k in place of $(\log n)^k$ changes the main term by a factor $1 + O(k/L)$ on $(N, 2N]$ and is harmless.

Lemma 16 (Gallagher [3])**.** *For each fixed k, $\displaystyle\sum_{\substack{D\subset[1,h]\\ |D|=k}} \mathfrak{S}(D) = \frac{h^k}{k!}(1+o(1))$ as $h\to\infty$.*

Theorem 17. *Assume* HL*. Then for every $\mu > 0$,*

$$\sup_{0\le h\le \mu L} \big|C_N(h) - e^{-2h/L}\big| \longrightarrow 0 \qquad (N\to\infty).$$

Proof. Step 1: factorial moments. Since $\mathcal{N}(n,h) = \sum_{d=1}^{h}\mathbf{1}_{\mathcal{P}}(n+d)$, we have exactly $\binom{\mathcal{N}(n,h)}{k} = \sum_{D\subset[1,h],|D|=k}\prod_{d\in D}\mathbf{1}_{\mathcal{P}}(n+d)$. Averaging over n and applying HL,

$$m_k(N,h) := \frac{1}{N}\sum_{N<n\le 2N}\binom{\mathcal{N}(n,h)}{k} = L^{-k}\Big(\sum_{\substack{D\subset[1,h]\\ |D|=k}}\mathfrak{S}(D) + \binom{h}{k}\varepsilon_k(N,h)\Big),$$

with $\sup_{h\le\mu L}|\varepsilon_k(N,h)| \to 0$. We claim $\sup_{0\le h\le\mu L}|m_k(N,h) - (h/L)^k/k!| \to 0$. Given $\delta > 0$, Lemma 16 gives H_0 such that the singular-series sum lies within a factor $1\pm\delta$ of $h^k/k!$ for $h \ge H_0$. For $H_0 \le h \le \mu L$ the difference is at most $\delta\mu^k/k! + \mu^k \sup|\varepsilon_k|$, using $\binom{h}{k} \le h^k/k!$. For $h < H_0$, both m_k and $(h/L)^k/k!$ are $O_{H_0,k}(L^{-k})$. Letting $N\to\infty$ and then $\delta\to 0$ proves the claim.

Step 2: from moments to parity. Suppose the theorem fails. Then there are $\delta > 0$, $N_i \to \infty$ and $0 \le h_i \le \mu L_i$ with $|C_{N_i}(h_i) - e^{-2h_i/L_i}| \ge \delta$. Passing to a subsequence, $h_i/L_i \to \lambda \in [0, \mu]$. Let Z_i be $\mathcal{N}(n, h_i)$ for n uniform in $(N_i, 2N_i]$. By Step 1 and continuity, $\mathbb{E}\binom{Z_i}{k} \to \lambda^k/k!$ for every k, and these are the factorial moments of a Poisson variable Z with mean λ. Ordinary moments are fixed linear combinations of factorial moments, so all moments of Z_i converge to those of Z. A Poisson law is determined by its moments, so by the method of moments ([2, §30]) $Z_i \to Z$ in distribution. For integer-valued variables this gives $\Pr(Z_i = j) \to \Pr(Z = j)$ for each j, and since the first moments are bounded the family is tight, so $\mathbb{E}f(Z_i) \to \mathbb{E}f(Z)$ for every bounded f. With $f(j) = (-1)^j$, $C_{N_i}(h_i) \to \mathbb{E}(-1)^Z = e^{-2\lambda}$, while $e^{-2h_i/L_i} \to e^{-2\lambda}$. This contradicts the choice of h_i. □

Corollary 18. *Assume* HL. *For $Y \ge 1$ let $V_N(Y) = \frac{1}{N}\sum_{N<a\le 2N}\big(\sum_{a<n\le a+Y}(-1)^{\pi(n)}\big)^2$. For every fixed $\mu > 0$, with $Y = \lfloor \mu L \rfloor$,*

$$\frac{V_N(Y)}{YL} \longrightarrow F(\mu) := 1 - \frac{1 - e^{-2\mu}}{2\mu}, \qquad \text{and} \qquad F(\mu) \to 1 \quad (\mu \to \infty).$$

Proof. Expanding the square, the pair $(a+u, a+v)$ with $1 \le u < v \le Y$ contributes the average of $(-1)^{\pi(n)}(-1)^{\pi(n+v-u)}$ over $n \in (N+u, 2N+u]$, which differs from $C_N(v-u)$ by $O(Y/N)$. Hence $V_N(Y) = Y + 2\sum_{h=1}^{Y-1}(Y-h)C_N(h) + O(Y^3/N)$. Divide by YL and insert Theorem 17; the uniform $o(1)$ error contributes at most $2\mu \cdot o(1)$, and the main term is a Riemann sum converging to $2\int_0^\mu (1 - t/\mu)e^{-2t}\,dt = F(\mu)$. □

A window of μL integers contains about μ gaps, so $V_N(Y)/(YL)$ is the variance of G over μ gaps per gap and per $\log^2$, the same normalization as S in Section 9.6. Corollary 18 therefore says that under Hardy–Littlewood the short-window constant is $F(\mu)$, and that $F(\mu) \to 1$ as the window grows. This proves, under HL, the iterated-limit form of the statement that the constant tends to 1; it does not by itself give the long-window statement, for which see the end of this section.

The convergence in N is slow. On $(3 \times 10^7, 6 \times 10^7]$, where $L \approx 17.56$:

μ	Y	$F(\mu)$	Poisson, this L	HL pairs	sieve ≤ 13	primes
0.5	9	0.368	0.370	0.361	0.358	0.358
1	18	0.568	0.560	0.531	0.518	0.516
2	35	0.755	0.724	0.663	0.637	0.630
4	70	0.875	0.832	0.743	0.707	0.695
8	141	0.938	0.888	0.784	0.741	0.730

Here "Poisson, this L" uses $C(h) = (1 - 2/L)^h$; "HL pairs" adds the pair-correlation correction from the singular series,

$$\log C(h) = h\log(1 - 2/L) + \frac{4}{L^2}\sum_{d<h}(h-d)\big(\mathfrak{S}(\{0,d\}) - 1\big);$$

and the last two columns, the model sieved by the primes up to 13 and the primes themselves, are measured. The primes lie below the limit $F(\mu)$ by an amount that grows with μ; about two thirds of the shortfall at $\mu = 8$ relative to the finite-L Poisson value is accounted for by pair correlations, and the sieve model, which carries correlations of all orders, comes within 0.011 of the primes. This is the finite-N behaviour the theorem permits; it is not a test of the theorem, which concerns $N \to \infty$.

11.2 A diffusive hypothesis implies existence and rates

Hypothesis D. There are constants K and k_0 such that for all integers $A \ge k_0$ and $1 \le B, m \le 3A$,

$$\frac{1}{B}\sum_{S=A+1}^{A+B}\big(G(S+m)-G(S)\big)^2 \;\le\; K\,m\log^2 A.$$

This is an upper-bound form of the diffusive behaviour measured in Section 9, where the left side divided by m is flat near $186 \approx S\log^2 p$ for $16 \le m \le 16384$. Under the heuristic of Section 9.6 it should hold with any $K > S$ for large A.

Theorem 19. *Assume* D *and fix a non-integral x. Then:*

(i) the series $\sum_k (-1)^k g_k\psi_k$ converges, so $T_M(x)$ is defined for every M, $N_M(x)$ converges, and $r_\infty(x)$ exists;

(ii) for all sufficiently large M_0,

$$\Big(\frac{1}{M_0}\sum_{M_0<M\le 2M_0} T_M(x)^2\Big)^{1/2} \le 10\sqrt{K}\,\frac{\log M_0}{\sqrt{M_0}};$$

(iii) $T_M(x) \ll_{x,K} M^{-0.475/3}(\log M)^{0.85}$ for every large M.

Proof. Write $\|f\| = \big(\frac{1}{M_0}\sum_{M_0<M\le 2M_0} f(M)^2\big)^{1/2}$ and assume $M_0 \ge 2(x+1)$, so that ψ_k is positive and decreasing for the k involved.

Windows. For $M \in (M_0, 2M_0]$ and $j \ge 0$ put $S_j = M - 1 + (2^j-1)M_0$ and $m_j = 2^j M_0$. The windows $(S_j, S_j+m_j]$ tile $[M,\infty)$. Let $W_j(M) = \sum_{k=S_j+1}^{S_j+m_j}(-1)^k g_k\psi_k$. With $H(\ell) = G(S_j+\ell) - G(S_j)$, summation by parts gives exactly

$$W_j = \psi_{S_j+m_j}H(m_j) + \sum_{\ell=1}^{m_j-1} H(\ell)\big(\psi_{S_j+\ell} - \psi_{S_j+\ell+1}\big).$$

As M runs over $(M_0, 2M_0]$, S_j runs over M_0 consecutive integers starting just above $A_j = 2^jM_0 - 1$, and $B = M_0$, $m_j \le 3A_j$. So Hypothesis D gives $\|H(\ell)\| \le \sqrt{K\ell}\log A_j$. The coefficients depend on M only through S_j and are bounded uniformly in M: $0 < \psi_{S+\ell} - \psi_{S+\ell+1} \le (A_j+\ell-x)^{-2}$ and $\psi_{S_j+m_j} \le (A_j - x)^{-1}$. So by Minkowski's inequality

$$\begin{aligned}\|W_j\| &\le \sqrt{K}\log A_j\Big(\frac{\sqrt{m_j}}{A_j-x} + \sqrt{m_j}\sum_{\ell\ge 1}\frac{1}{(A_j+\ell-x)^2}\Big)\\ &\le \frac{2\sqrt{K}\sqrt{m_j}\log A_j}{A_j-x} \le \frac{4\sqrt{K}(\log M_0 + j\log 2)}{\sqrt{2^jM_0}},\end{aligned}$$

using $A_j - x \ge 2^{j-1}M_0$.

(ii). Summing over j,

$$\sum_j \|W_j\| \le \frac{4\sqrt{K}(3.42\log M_0 + 5.72)}{\sqrt{M_0}} \le \frac{20\sqrt{K}\log M_0}{\sqrt{M_0}} \qquad (\log M_0 \ge 3.7).$$

Since $T_M = \frac{1}{2}\sum_j W_j(M)$, (ii) follows from Minkowski's inequality once the series is known to converge, which is (i).

(i). On the finite set $(M_0, 2M_0]$, $|f(M)| \le \sqrt{M_0}\|f\|$, so $\sum_j |W_j(M)| < \infty$ for each M: the series converges along the window endpoints, to a value T^*_M. These values are consistent. For $M < M'$ in the same range, the endpoints of corresponding windows differ by the fixed amount $M' - M$; for M in $(M_0, 2M_0]$ and M' in $(2M_0, 4M_0]$, window $j+1$ of the first system and window j of the second have endpoints differing by the fixed amount $M' - M - M_0$. In both cases the partial sums along the two systems differ by a block of boundedly many terms at height tending to infinity, and each term is $g_k\psi_k \ll k^{-0.475}(\log k)^{0.525} \to 0$ by the Baker–Harman–Pintz bound $g_k \ll p_k^{0.525}$ [1]. Hence $Q := \sum_{k<M}(-1)^k g_k\psi_k + T^*_M$ does not depend on M, and the series converges, to Q, if and only if $T^*_M \to 0$. Now $|T^*_M - T^*_{M'}| \le \frac{1}{2}|M - M'|\eta$ with $\eta = \max_{k \le 2M_0 + |M-M'|} g_k\psi_k \ll M_0^{-0.475}(\log M_0)^{0.525}$ for k in the relevant range. If $|T^*_M| \ge \varepsilon$ for some $M \in (M_0, 2M_0]$, then $|T^*_{M'}| \ge \varepsilon/2$ for all M' in that range within distance $R = \lfloor \varepsilon/\eta \rfloor$ of M, and there are at least $\min(R+1, M_0)$ of them. Since $T^*_M = \frac{1}{2}\sum_j W_j(M)$, the bound on $\sum_j \|W_j\|$ gives $\|T^*\| \le 10\sqrt{K}\log M_0/\sqrt{M_0}$ without using (i), so $\min(R+1, M_0)\varepsilon^2/4 \le 100K\log^2 M_0$. If $R + 1 \le M_0$ this gives $\varepsilon^3 \ll K\eta\log^2 M_0 \ll M_0^{-0.475}(\log M_0)^{2.525}$; otherwise $\varepsilon \ll \log M_0/\sqrt{M_0}$, which is smaller. Either way $\sup_{M_0 < M \le 2M_0} |T^*_M| \ll M_0^{-0.475/3}(\log M_0)^{0.842} \to 0$, which proves (i) and (iii). The analytic part of the tail converges by Proposition 5, the denominator D_M converges unconditionally, and $D_\infty(x) \ne 0$, so N_M and r_M converge. □

Corollary 20. *Under Hypothesis* D, $r_M(x) - r_\infty(x) = O(|T_M(x)|) + O_x(\log M/M)$. *Hence the root mean square of* $r_M - r_\infty$ *over* $M \in (M_0, 2M_0]$ *is* $\ll_x \sqrt{K}\log M_0/\sqrt{M_0}$, *and* $r_M(x) - r_\infty(x) \ll_{x,K} M^{-0.158}(\log M)^{0.85}$ *for every large* M.

Proof. As in the proof of Proposition 7: $N_\infty - N_M$ is the arithmetic tail T_M plus an analytic tail of size $O(\log M/M)$ by Proposition 5, $D_\infty - D_M = O(\log M/M)$, and D_M is bounded away from 0 for large M. Then apply Theorem 19. □

The mean-square rate $\log M/\sqrt{M}$ is the rate predicted heuristically in Section 9.6, now proved under D; in mean square it also removes the $\sqrt{\log\log M}$ factor of Corollary 10. The pointwise exponent 0.158 is weaker than the expected $1/2$ because a mean-square hypothesis controls typical M, and passing to every M uses only the smallness of single gaps.

11.3 A converse: existence of r_∞ forces $G(k) = o(k)$

Proposition 21. *If* $r_M(x_0)$ *converges as* $M \to \infty$ *for a single non-integral* x_0, *then* $G(k) = o(k)$; *equivalently,* $\sum_{2 \le n < p_{k+1}} (-1)^{\pi(n)} = o(k)$.

Proof. $D_M(x_0)$ converges to $D_\infty(x_0) \ne 0$, so $N_M(x_0)$ converges. By Lemma 4 and (21), N_M is a partial sum of $-\frac{1}{2}\sum_k (-1)^k \Delta a_k$, whose analytic part converges by Proposition 5. So $\sum_k (-1)^k g_k/(k + 1 - x_0)$ converges. The denominators $k + 1 - x_0$ are positive and increase to infinity for $k > x_0$, so Kronecker's lemma gives $\sum_{j \le k}(-1)^j g_j = o(k)$. The second form is Corollary 15. □

The existence of r_∞ is therefore not a statement about interpolation alone: it implies that the parity of $\pi(n)$, weighted along the integers, has mean zero at a rate better than the trivial one. Section 9 records that the trivial bound $|G(k)| \ll k\log k$ has not been improved here, and we do not know of an unconditional proof of $G(k) = o(k)$. If there is none, the existence of r_∞ is at present conditional, on the hypothesis of Corollary 8 through Proposition 6, or on Hypothesis D through Theorem 19, and supported numerically.

11.4 What is proved and what is not

Proved unconditionally. Propositions 11, 12 and 14, Corollary 15, and Proposition 21.

Proved under HL *(Gallagher's uniform form).* Poisson parity correlations on the scale $\log N$, uniformly for $h \le \mu \log N$ (Theorem 17), and the short-window variance law with constant $F(\mu) \to 1$ (Corollary 18).

Proved under D. Existence of r_∞, the mean-square rate $\log M/\sqrt{M}$, and the pointwise rate $M^{-0.158}$ up to logarithms (Theorem 19, Corollary 20).

Not proved. That HL implies D. Theorem 17 controls windows of length $\mu \log N$ for each fixed μ, because each fixed k in HL controls only k-point correlations on that scale. Hypothesis D needs windows of every length up to a constant times the height, which requires the parity of π to decorrelate over distances much larger than $\log N$. That would need HL with uniformity in k growing with N, of the kind Montgomery and Soundararajan [6] assume for moments of primes in short intervals, or a separate argument. This is the missing link between the two halves of the section.

12 Richardson extrapolation to arbitrary order

By Proposition 7 the error splits as $r_M = r_\infty + (\text{analytic}) + (\text{arithmetic})$, and only the analytic part admits an expansion $a_1/M + a_2/M^2 + \cdots$. Extrapolation in $1/M$ can therefore cancel that part and nothing else, which is worth keeping in view when reading the table below. Sampling at $M_i = N \cdot \rho^i$ for $i = 0, \dots, q$ and writing $t_i = 1/M_i$, evaluation of the degree-q interpolating polynomial in t at $t = 0$ gives Lagrange coefficients

$$\prod_{j \ne i} \frac{-t_j}{t_i - t_j} = \prod_{j \ne i} \frac{-\rho^{-j}}{\rho^{-i} - \rho^{-j}} = \prod_{j \ne i} \frac{1}{1 - \rho^{j-i}},$$

so that for any order

$$R_q(x) = \sum_{i=0}^{q} \lambda_i \, r_{N\rho^i}(x), \qquad \lambda_i = \prod_{\substack{j=0 \\ j \ne i}}^{q} \frac{1}{1 - \rho^{j-i}}. \tag{28}$$

With $\rho = 2$ the first few coefficient sets are (1), $(-1, 2)$, $(\frac{1}{3}, -2, \frac{8}{3})$ and $(-\frac{1}{21}, \frac{2}{3}, -\frac{8}{3}, \frac{64}{21})$, each summing to 1.

Measured against the limit, averaging |error| over $x = 10.5, 25.5, 60.5, 125.5$:

primes used	N	$q = 0$	$q = 1$	$q = 2$	$q = 3$
1024	128	0.12144	0.06042	0.02149	0.00206
1000	250	0.08055	0.03586	**0.00082**	0.00194
2000	500	0.05820	0.00841	0.00179	0.04374
16000	1000	0.03330	0.00121	0.03805	0.09699

The reference limit here is the block mean of the earlier version of Section 13, from 1.03×10^6 primes, which differs from the value computed from 6.84×10^6 primes by about 3×10^{-4} at the half-integers; differences below a few times 10^{-4} in this table are not resolved.

Order 2 with 1000 primes reaches 8×10^{-4}, a hundredfold gain. Beyond order 3 the extrapolation degrades, and the last row shows the counterintuitive consequence that with 16000 primes $q = 2$ is worse than $q = 1$.

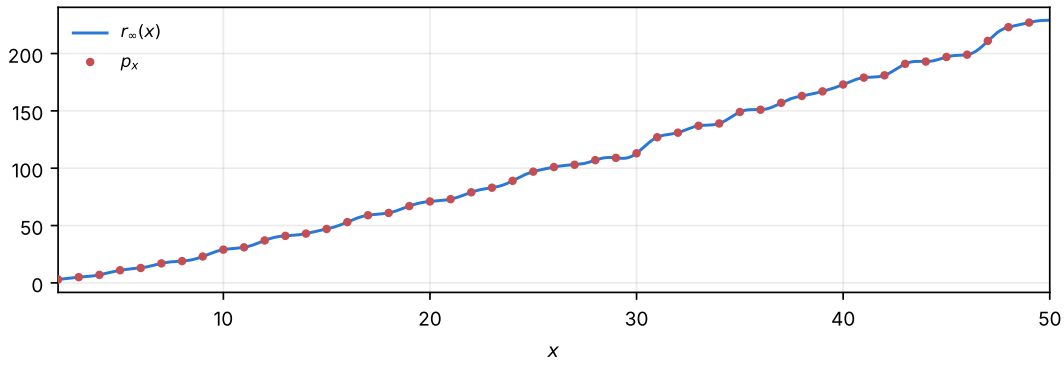


Figure 6: The limit function r_∞ on $2 \le x \le 50$, computed from 1.03×10^6 primes, with the primes marked; at this scale more primes make no visible difference. It interpolates every p_k exactly and involves no free parameters.

Section 8 accounts for this. Richardson removes the analytic part, which Proposition 5 places at $O(\log M/M)$ and which the table at the end of that section shows sitting near 3×10^{-5} by $M = 10^5$. What remains is the arithmetic part, which is not a power series in $1/M$ at all and which the same table shows at 2×10^{-2} there. Once the analytic part has been cancelled, adding orders can only take differences of the arithmetic part, and differencing a random-walk-like quantity amplifies it by the size of the extrapolation coefficients, which for $\rho = 2$ grow like 2^q. Two facts follow, both visible in the table: the gain saturates near 10^{-3} rather than continuing, and more primes do not help once you are at the floor, because the floor is set by G and not by M alone.

13 Results

The limit values below are computed from all 6 841 648 primes below 1.2×10^8. The stated uncertainty is half the spread of r_M over $M \in [3.42 \times 10^6, 6.84 \times 10^6]$, which is 1.1×10^{-3} at every abscissa, as the Nyquist form of the error predicts.

x	$r_\infty(x)$	bracketing primes
2.5	4.1307 ± 0.0011	3, 5
3.5	5.6748 ± 0.0011	5, 7
10.5	30.0213 ± 0.0011	29, 31
20.5	71.7034 ± 0.0011	71, 73
60.5	281.7371 ± 0.0011	281, 283
125.5	696.0030 ± 0.0011	691, 701
200.5	1226.2469 ± 0.0011	1223, 1229

An earlier version of these notes listed these values to six decimals, as block means over $M \in [3 \times 10^5, 1.03 \times 10^6]$ whose own spread was 5.4×10^{-3}. Those values agree with the table above within its uncertainty, but their last three digits were not significant. Richardson extrapolation in $1/M$ from the pair $M = 10^5, 10^6$ gives 281.7369 at $x = 60.5$, also within the uncertainty; since the arithmetic part of the error is not a power series in $1/M$, that agreement is a consistency check, not an independent confirmation at the level of 10^{-4}. Figure 6 plots the limit function.

Two consistency checks are worth recording. The centred-window construction of Section 4, using only 250 primes, gave 695.840 at $x = 125.5$, which differs from the limit by 0.16; that is

inside its own $O(h^{-1/2})$ error bar. And the limit is well behaved at the left edge, where every window-based method fails for want of room to centre: $r_\infty(2.5) = 4.131$ and $r_\infty(3.5) = 5.675$ both lie inside their bracketing pairs.

14 What is and is not established

Established. The Newton series (1) diverges at every non-integer x, with the mechanism and rate of Section 2. Recentring gives convergence at rate $O(h^{-1/2})$, by the cancellation (7). In the barycentric $d = 1$ scheme the denominator converges, in closed form; the numerator's partial sums are, by Lemma 4, identically a half-sum of first differences, which is why the endpoint halving matters. The scheme is exactly a ratio of cardinal series (Proposition 11), and its error is exactly $\sin(\pi x)$ times a slowly varying factor (Proposition 12); numerically that factor is constant across eight abscissas to five decimals. Proposition 5 bounds the analytic part of the error by $O(\log M/M)$ unconditionally, using only $p_k \ll k \log k$; Proposition 6 bounds the arithmetic part by $O(\Phi(M)/M)$ for any regular majorant Φ of $|G|$. The alternating gap sum is exactly $\sum(-1)^{\pi(n)}$ over the integers (Corollary 15), and the existence of r_∞ forces $G(k) = o(k)$ (Proposition 21).

Established conditionally. Under Hypothesis D, a diffusive bound on increments of G, the limit r_∞ exists, with root-mean-square rate $\log M/\sqrt{M}$ and pointwise rate $M^{-0.158}$ up to logarithms (Theorem 19). Under the Hardy–Littlewood conjecture in Gallagher's form, the parity correlations of π are Poisson on the scale $\log N$ and the short-window variance constant tends to 1 (Theorem 17, Corollary 18).

Supported numerically. The limit values of Section 13, to about $\pm 1.1 \times 10^{-3}$. Diffusive behaviour of G with no drift over six decades, with long-run step variance about five percent above $\mathrm{Var}(g)$. Agreement of the constant S with the Hardy–Littlewood sieve models, and disagreement with the Cramér model by about seven standard errors. A typical size of the arithmetic tail matching (26).

Refuted. $c_M = O(1/M)$, asserted in an earlier version of these notes on the strength of a pointwise log-log fit. By Proposition 7 it would force $G(k) = O(\log k)$, and $|G(2^{18})| = 1.23 \times 10^4$ against $\log 2^{18} \approx 12.5$. The fit that produced it was computed with an estimator that is invalid for a quantity containing a random-walk term.

Not established. The existence of r_∞ unconditionally. An earlier version listed it as established; by Proposition 21 it implies $G(k) = o(k)$, which we cannot prove, and unconditionally we have only $c_M \ll \log M$, which does not even give $c_M \to 0$. The true size of $G(k)$: Section 9 reduces it to square-root cancellation in $\sum_{j \le k}(-1)^j(g_j - \log p_j)$, equivalently in the alternating sum of the primes, and shows that the Riemann Hypothesis does not reach it, since RH controls the unsigned sum $A + B$ while the question is about $A - B$. That the Hardy–Littlewood conjecture implies Hypothesis D; Section 11 explains what is missing. There is no uniqueness theorem behind any of this. The construction is parameter-free once $d = 1$ and $M \to \infty$ are fixed, but that makes it a canonical recipe, not a proven canonical function; a different equally parameter-free scheme could land elsewhere. Carlson's theorem cannot arbitrate, because r_∞ is a limit of rational functions with poles in $\mathbb{C}$, not an entire function of exponential type, so the type-π uniqueness machinery does not apply to it. And the extrapolation floor near 10^{-3} is an empirical observation from four abscissas, not a theorem.

A recurring number. At 250 primes, six unrelated routes gave the same uncertainty of roughly 0.1: the truncation residual of the centred window, the switch-point jump (8), the partition-of-unity blend envelope, the disagreement between the barycentric and window constructions, the sensitivity to the centring offset, and the dependence on M. That is not a defect of any one method. It is how much the interpolated value is determined by that much data.

A Implementation

Desmos. With the prime list `p` and $M = \text{length}(p)$, $K = [1, \dots, M]$:

$$u(n) = \{K \le n : 1, 0\}\,(-1)^K\big(2 - \{K = 1 : 1, 0\} - \{K = n : 1, 0\}\big),$$
$$g(x, n) = \text{total}\left(\tfrac{u(n)\,p}{x-K}\right) \Big/ \text{total}\left(\tfrac{u(n)}{x-K}\right),$$
$$L(i) = \prod_{j=0}^{q} \big\{j = i : 1,\ (1 - 2^{j-i})^{-1}\big\}, \qquad R(x) = \sum_{i=0}^{q} L(i)\, g\big(x,\, N \cdot 2^i\big),$$

with $N = \text{floor}(M/2^q)$. Note that $g(x, n)$ is meaningful only for x inside $[1, n]$, so $R(x)$ interpolates the primes only for $x \le N$; taking N too small relative to the plotted range produces errors in the hundreds rather than the thousandths.

Numerics. The difference tables were computed in exact integer arithmetic and evaluated with `mpmath` at 50–80 decimal digits, which is necessary: the centred sum at $h = 248$ cancels about 150 digits, and double precision fails there silently, returning a plausible but wrong curve. The barycentric computations are well conditioned and were done in double precision, with r_M for all M obtained in one pass by cumulative sums over a sieve of 1 031 130 primes below 1.6×10^7. Sections 9.6, 10, 11 and 13, and Figures 2 and 3, use all 6 841 648 primes below 1.2×10^8; the random-model comparisons use independent simulated sequences over the same ranges.

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